

An exact nonlinear dual formula on the two-site birth–death chain

A substantial partial answer to Open Question 4 of arXiv:math/0101227

Autonomous mathematical discovery run `fa_banach_001`, lane 8

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Status. Candidate substantial partial result, likely valid. For every $p > 2$ we prove a complete nonlinear sup–inf/inf–sup formula for the sharp ℓ^p Sobolev constant of the first nontrivial Dirichlet birth–death chain. This is the Sobolev partner of a Nash inequality, but it is not the centered ergodic Nash or logarithmic-Sobolev constant asked for in full by the source.

1 Source question and later-literature boundary

Chen asks for a one-dimensional Nash criterion and, separately, for a variational formula in the completed dual form of his Theorem 4.1(2) for a Nash or logarithmic-Sobolev inequality [1, p. 6, Open Problem 3 and Open Question 4]. The exact source passage is reproduced below.



Table 5.1

Here, “(*) & ...” means that one requires the uniqueness condition in the first line plus the condition “...”. The “(ε)” in the last line means that there is still a small gap from being necessary.

Open problem 3. *Find out a criterion for the Nash inequality in dimension one.*

In view of Theorem 4.1, it is natural to ask the following

Open question 4. *Does there exist a variational formula in the form of part (2) of Theorem 4.1 for the logarithmic Sobolev or Nash inequality?*

DIFFUSION PROCESSES ON $[0, \infty)$

Figure 1: Open Problem 3 and Open Question 4 in the source PDF, page 6.

Mao subsequently gave if-and-only-if Nash criteria for birth–death processes and diffusions on the line, explicitly identifying Chen’s question [2]. Thus Problem 3 is already answered. Chen later extended the Poincaré formulas to Banach/Orlicz norms [3, 4]. Those results give exact criteria, explicit constant-factor bounds, and inf–sup formulas for *upper bounds*; they do not state the exact dual equality requested for the centered Nash or logarithmic-Sobolev constant. The result below isolates a setting in which the exact nonlinear equality can be recovered.

2 Proof intuition

The derivative of an L^p Sobolev quotient is nonlinear: its source is u^{p-1} , not u . On a one-dimensional Dirichlet chain, summing the Euler–Lagrange equation from a site to the Neumann endpoint converts that source into a positive tail. Dividing the tail by the edge flux gives a nonlinear analogue of

Chen's I -functional. A summation-by-parts identity proves the sup–inf half on every finite chain. For two sites, the two tail functionals are respectively strictly decreasing and strictly increasing in the only projective coordinate, so their crossing proves the dual inf–sup half as well.

3 A general exact sup–inf half-formula

Let $n \geq 1$, let masses $\mu_i > 0$ and conductances $c_i > 0$, and put $f_0 = 0$. For $p > 2$, define

$$\|f\|_{p,\mu} = \left(\sum_{i=1}^n \mu_i |f_i|^p \right)^{1/p}, \quad \mathcal{E}(f) = \sum_{i=1}^n c_i (f_i - f_{i-1})^2,$$

and the sharp Dirichlet Sobolev constant

$$A_p = \max_{f \neq 0} \frac{\|f\|_{p,\mu}^2}{\mathcal{E}(f)}.$$

For $f_0 = 0 < f_1 < \dots < f_n$, set

$$J_i(f) = \frac{\sum_{j=i}^n \mu_j f_j^{p-1}}{c_i (f_i - f_{i-1}) \|f\|_{p,\mu}^{p-2}}. \quad (1)$$

Lemma 1. *For every finite Dirichlet birth–death chain and every $p > 2$,*

$$A_p = \sup_{0=f_0 < f_1 < \dots < f_n} \min_{1 \leq i \leq n} J_i(f).$$

Proof. Let $m = \min_i J_i(f)$. Multiply the defining inequality for m by $f_i - f_{i-1}$ and sum over i . Reversing the finite sums gives

$$\begin{aligned} m \|f\|_{p,\mu}^{p-2} \mathcal{E}(f) &\leq \sum_{i=1}^n (f_i - f_{i-1}) \sum_{j=i}^n \mu_j f_j^{p-1} \\ &= \sum_{j=1}^n \mu_j f_j^{p-1} \sum_{i=1}^j (f_i - f_{i-1}) = \|f\|_{p,\mu}^p. \end{aligned}$$

Hence $m \leq \|f\|_{p,\mu}^2 / \mathcal{E}(f) \leq A_p$.

The maximum defining A_p exists. Replacing an optimizer by its absolute value cannot increase the energy. Its Euler equation and the positive inverse of the Dirichlet stiffness matrix then make it strictly positive; summing the Euler equation from i to n makes it strictly increasing. For such an optimizer u ,

$$c_i (u_i - u_{i-1}) = \frac{\mathcal{E}(u)}{\|u\|_{p,\mu}^p} \sum_{j=i}^n \mu_j u_j^{p-1}.$$

Thus every $J_i(u) = \|u\|_{p,\mu}^2 / \mathcal{E}(u) = A_p$, proving the reverse inequality and the lemma. $\square$

4 The complete two-site dual formula

Take $n = 2$, $\mu_1 = \mu_2 = c_1 = c_2 = 1$. Thus

$$\mathcal{E}(x) = x_1^2 + (x_2 - x_1)^2, \quad A_p = \max_{x \neq 0} \frac{(|x_1|^p + |x_2|^p)^{2/p}}{x_1^2 + (x_2 - x_1)^2}.$$

For $0 < s < 1$, put

$$Q_p(s) = \frac{(1 + s^p)^{2/p}}{s^2 + (1 - s)^2}, \quad (2)$$

$$J_1(s) = \frac{1 + s^{p-1}}{s(1 + s^p)^{(p-2)/p}}, \quad J_2(s) = \frac{1}{(1 - s)(1 + s^p)^{(p-2)/p}}. \quad (3)$$

Theorem 1. *For every $p > 2$, there is a unique $s_p \in (0, 1)$ satisfying*

$$1 - 2s_p + s_p^{p-1} - s_p^p = 0. \quad (4)$$

The sharp two-site constant has the completed nonlinear dual formula

$$A_p = \sup_{0 < s < 1} \min\{J_1(s), J_2(s)\} = \inf_{0 < s < 1} \max\{J_1(s), J_2(s)\} = Q_p(s_p) = J_1(s_p) = J_2(s_p). \quad (5)$$

Proof. Let

$$H_p(s) = 1 - (p - 2)s^{p-1} + (p - 1)s^p.$$

Its minimum on $[0, 1]$ occurs at $r = (p - 2)/p$ and equals $1 - r^p > 0$. Direct logarithmic differentiation of (3) gives

$$J_1'(s) < 0, \quad J_2'(s) > 0,$$

with the two derivative numerators equal to $-H_p(s)$ and $H_p(s)$, respectively, after multiplication by positive denominators. Moreover, $J_1(s) = J_2(s)$ is exactly equation (4). Since J_1 starts above J_2 at 0 and ends below it at 1, there is one crossing s_p . The minimum of the maximum and the maximum of the minimum therefore both occur at this crossing.

Replacing (x_1, x_2) by $(|x_1|, |x_2|)$ cannot increase the denominator of the Sobolev quotient. After scaling $x_2 = 1$, its relevant projective coordinate is $s = x_1/x_2 \geq 0$. On $(0, \infty)$,

$$\frac{Q_p'(s)}{Q_p(s)} = \frac{2(1 - 2s + s^{p-1} - s^p)}{(1 + s^p)(s^2 + (1 - s)^2)}.$$

The sign change just proved shows that Q_p increases up to s_p and decreases thereafter (for $s \geq 1$ the displayed numerator is negative). Thus $A_p = Q_p(s_p)$. Finally, equation (4) implies

$$s_p^2 + (1 - s_p)^2 = (1 - s_p)(1 + s_p^p),$$

so $Q_p(s_p) = J_2(s_p) = J_1(s_p)$. This proves (5). □

5 The simpler Banach tail formula is strictly nonsharp

The Banach-space upper formula in Chen's later theory replaces the nonlinear tail f^{p-1} by the norm of an ordinary tail. At $p = 4$, the relevant Banach norm on f^2 is ℓ^2 . Its direct two-site tail functional is

$$U_4 = \inf_{t > 1} \max \left\{ \sqrt{1 + t^2}, \frac{t}{t - 1} \right\}. \quad (6)$$

Theorem 2. *For the two-site chain,*

$$A_4 < 2.08 < 2.13 < U_4.$$

In particular, the upper formula (6) cannot be upgraded to an exact identity even in this smallest nontrivial model.

Proof. The branches in (6) are strictly increasing and decreasing, respectively. They meet at the unique root t_* of

$$t^4 - 2t^3 + t^2 - 2t + 1 = 0.$$

Exact rational evaluation gives $1.883 < t_* < 1.884$, hence

$$U_4^2 > 1 + (1.883)^2 = 4.545689 > (2.13)^2.$$

Equation (4) at $p = 4$ is equivalent to $s^4 - s^3 + 2s - 1 = 0$. Exact evaluation gives $0.535 < s_4 < 0.536$. On this interval the numerator $1 + s^4$ and denominator $2s^2 - 2s + 1$ are increasing, so

$$A_4^2 < \frac{1 + (0.536)^4}{(2(0.535)^2 - 2(0.535) + 1)^2} = \frac{270634747904}{63114000625} < (2.08)^2.$$

All signs and rational comparisons are independently reproduced by the supplied verifier. □

6 Verification, novelty bounds, and limitations

Run `code/verify_two_site.py` from the repository root. It checks the $p = 4$ derivative factorizations symbolically, certifies both rational root brackets and the strict 0.05 separation, and evaluates the general formula at five exponents with 60-decimal arithmetic. The proof for arbitrary $p > 2$ is algebraic and does not depend on those numerical checks.

The bounded novelty search covered the run registry and attempt indexes, Chen’s 2001 source, Mao’s 2002 Nash paper, Chen’s 2002/2003 Banach-space and birth–death papers, the later monograph chapter summarizing those results, and web/arXiv searches combining “Collatz–Wielandt,” “Nash,” “Sobolev constant,” “birth–death,” and “variational formula.” General nonlinear Perron–Frobenius theory is known, but no source stating formulas (5) or the strict comparison in Theorem 2 was found. This is bounded evidence only; the specialization may be folklore.

The result is deliberately scoped. The centered Neumann Nash quotient has an Euler source containing both f and a constant from its L^1 factor; the logarithmic-Sobolev source contains $f \log(f^2/\|f\|_2^2)$. Both change sign, destroying the positive-tail ordering used here. A reviewer should verify that this distinction is maintained and that the packet is not read as a solution of the full source question.

References

- [1] M.-F. Chen, *Explicit criteria for several types of ergodicity*, Chinese J. Appl. Probab. Statist. 17 (2001), 1–8; [arXiv:math/0101227](#).
- [2] Y.-H. Mao, *Nash inequalities for Markov processes in dimension one*, Acta Math. Sin. (Engl. Ser.) 18 (2002), 147–156, [doi:10.1007/s101140100128](#).
- [3] M.-F. Chen, *Variational formulas of Poincaré-type inequalities in Banach spaces of functions on the line*, Acta Math. Sin. (Engl. Ser.) 18 (2002), 417–436, [doi:10.1007/s10114-002-0176-8](#).
- [4] M.-F. Chen, *Variational formulas of Poincaré-type inequalities for birth–death processes*, Acta Math. Sin. (Engl. Ser.) 19 (2003), 625–644, [doi:10.1007/s10114-003-0308-9](#).